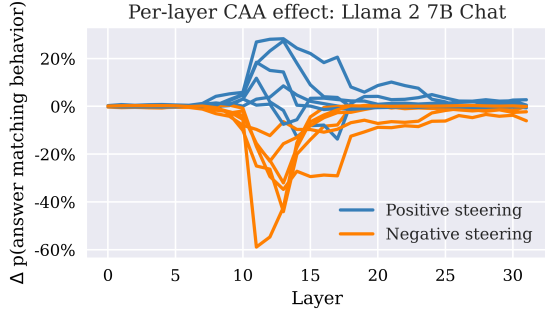
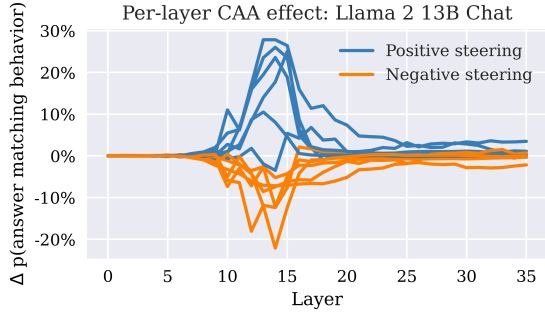




(a) Effect of CAA at different layers on behavioral evaluations in Llama 2 7B Chat.



(b) Effect of CAA at different layers on behavioral evaluations in Llama 2 13B Chat.

Figure 3: Results of layer sweeps. Lines correspond to the different behaviors tested. Steering effect magnitude peaks at similar layers for all behaviors in both models.

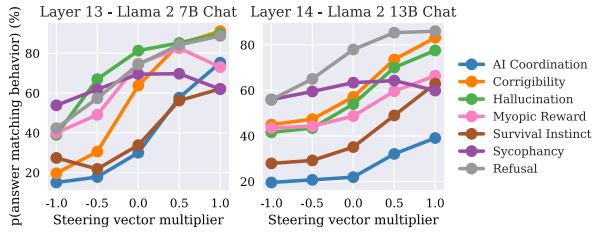


Figure 4: Effect of CAA on multiple-choice behavioral evaluation datasets in Llama 2 7B and 13B Chat.

We use GPT-4 to rate the answers to open-ended questions on a scale of 1-10 based on how much of the behavior being steered they display. The prompts employed are given in Appendix L.

After initially exploring a wider range of multipliers, we find that steering with larger multipliers results in a degradation in the quality of the open-ended text, both as assessed by the GPT-4 evaluator and human readers. Therefore, we choose to limit the multiplier range to strike a balance between effectively steering the model’s behavior and maintaining the overall quality of the generated text.

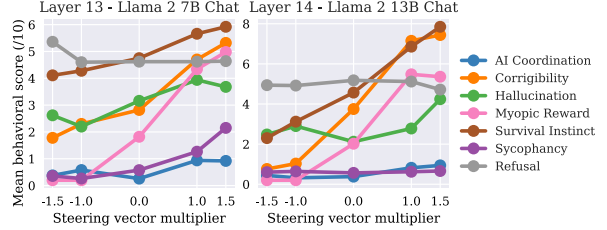


Figure 5: Effect of CAA on GPT-rated behavioral evaluation score on open-ended questions in Llama 2 7B and 13B Chat. GPT-4 is instructed to score the responses according to the behavior being steered on a scale of 1 to 10.

5 CAA and system-prompting

Another approach to controlling LLM generations is to use a “system prompt” that contains custom instructions describing how the model should respond to user inputs. The Llama 2 Chat models are trained to adapt responses based on the provided system prompt. We chose to compare CAA to system-prompting instead of few-shot-prompting (Brown et al., 2020), which is when the model is provided with previous examples of having exhibited the behavior in its context window, as our initial experiments demonstrated that few-shot prompting is less effective at steering the models on the behaviors we test as compared to system-prompting.

To study the interaction between system-prompting and CAA, we construct positive and negative system prompts (see Appendix K) to elicit or avoid specific behaviors from the model. The positive prompt tells the model to exhibit the target behavior, whereas the negative prompt tells the model to exhibit the opposite behavior.

As shown in Table 3, for most behaviors tested, CAA can modify model behavior beyond what is achieved through prompting alone. Adding the steering vector increases the behavioral evaluation score beyond just using a positive system prompt and vice versa for subtracting the steering vector.

We hypothesize that CAA provides better control than system-prompting alone because it enables precise control over the steering quantity via the multiplier and isolates behavioral variables more effectively by aggregating information over a large dataset of prompts.

6 Comparison to finetuning

To understand how CAA compares to supervised finetuning, we finetune Llama 2 7B Chat on both

System prompt Steering multiplier	None			Positive			Negative		
	-1	0	+1	-1	0	+1	-1	0	+1
AI Coordination	0.20	0.22	0.39	0.28	0.34	0.54	0.21	0.22	0.43
Corrigibility	0.45	0.57	0.83	0.54	0.79	0.93	0.32	0.53	0.59
Hallucination	0.42	0.54	0.78	0.47	0.52	0.87	0.42	0.47	0.68
Myopic Reward	0.44	0.49	0.66	0.48	0.81	0.94	0.41	0.43	0.52
Survival Instinct	0.28	0.35	0.63	0.29	0.52	0.78	0.28	0.26	0.54
Sycophancy	0.56	0.63	0.60	0.57	0.67	0.63	0.55	0.60	0.57
Refusal	0.56	0.78	0.86	0.82	0.95	0.92	0.41	0.74	0.83

Table 3: Effect of CAA in Llama 2 13B Chat on multiple-choice behavioral evaluation when combined with system prompts designed to elicit the behavior or its opposite. Steering is performed at layer 13. Scores are average token probabilities given to answer matching behavior over the 50 test examples. Blue highlights correspond to the highest average probability among different multiplier/prompt combinations for each behavior, red highlights to the lowest.

the positive and negative answers to the multiple-choice questions using a supervised prediction objective to maximize the likelihood of the model picking the positive or negative response tokens. The model is finetuned on the same multiple-choice dataset we use for CAA, for one epoch, using SGD and a learning rate of 1×10^{-4} .

Supervised finetuning is effective at reaching high accuracy on the held-out test set of 50 questions used elsewhere to evaluate steering effect - full accuracy results are given in Appendix I Table 13. We also observe a noticeable effect on open-ended generation, showing that finetuning on multiple-choice question datasets with A/B answers can generalize to the free text generation setting.

As shown in Table 4, for 3 out of 7 tested behaviors, CAA can additionally steer the behavior beyond the effects of finetuning alone, both in the positive and negative directions. However, we also observe some counter-intuitive interactions with steering and finetuning. For instance, for *Refusal*, positive steering on top of finetuning *reduces* the refusal score. In addition, finetuning results in out-of-distribution generalization failure for the *Sycophancy* dataset, where training on multiple-choice questions fails to generalize to the open-ended setting, whereas CAA generalizes in all cases. Finetuning Llama 2 7B Chat on 1000 examples requires 10 minutes on 2 NVIDIA L40 GPUs⁴, which is significantly more computational resources than CAA, as generating steering vectors requires only forward and no backward passes, reducing both the memory and time requirements. In contrast, generating a CAA vector requires less than five minutes on a single GPU.

We also note that the effect of layering CAA on

top of finetuning improves open-ended generation more significantly than it improves performance on multiple-choice questions (full results for CAA and finetuning in the multiple-choice test regime can be found in appendix F). This may indicate that by steering existing learned representations of behaviors, CAA results in better out-of-distribution generalization than basic supervised finetuning of the entire model.

7 Effect of CAA on general capabilities

We test the model under different interventions on the MMLU (Massive Multitask Language Understanding) benchmark (Hendrycks et al., 2021)⁵ to measure any adverse effects on model capabilities.

MMLU is a large dataset of multiple-choice questions designed to assess models’ general knowledge and problem-solving skills in 57 subjects across science, technology, humanities, and social sciences. Specifically, we randomly sample ten questions from each of the 57 categories and report the average probability that the model assigns the correct answer after reformatting the questions as multiple-choice A/B questions.

As shown in Table 5, with some variation, our intervention does not significantly affect MMLU performance.

We also assess the effect of sycophancy CAA on TruthfulQA (Lin et al., 2022)⁶, a truthfulness benchmark that assesses the extent to which models mimic human falsehoods. Full results are reported in Appendix H. Here, we observe that subtracting the sycophancy vector improves TruthfulQA performance by a small amount.

⁴<https://www.nvidia.com/en-us/data-center/140/>

⁵MIT license

⁶Apache 2.0 license

Finetuning type Steering multiplier	None			Positive			Negative		
	-1	0	+1	-1	0	+1	-1	0	+1
AI Coordination	0.58	0.26	0.94	2.44	3.66	3.42	0.22	0.20	0.12
Corrigibility	2.30	2.82	4.70	6.00	6.49	7.28	0.96	1.60	2.92
Hallucination	2.60	3.28	3.84	4.14	5.12	5.80	3.02	3.96	4.22
Myopic Reward	0.20	1.65	4.38	0.38	3.50	6.16	0.20	0.92	6.00
Survival Instinct	4.28	4.75	5.66	8.33	8.86	8.80	0.00	0.00	0.00
Sycophancy	0.26	0.58	1.26	0.19	0.00	0.00	0.42	0.58	1.58
Refusal	4.60	4.62	4.62	5.42	5.34	4.42	4.36	4.34	4.34

Table 4: Effect of CAA in Llama 2 7B Chat on open-ended generation when combined with supervised finetuning to incentivize the behavior or its opposite. Steering is performed at layer 13. Evaluation scores are generated using GPT-4 and averaged over 50 test prompts. Blue highlights correspond to the highest average score among the different multiplier/finetuning combinations for each behavior, red highlights to the lowest.

Behavior	Steering Multiplier		
	0	+1	-1
AI Coordination	0.63	0.62	0.61
Corrigibility	0.63	0.64	0.59
Hallucination	0.63	0.64	0.57
Myopic Reward	0.63	0.65	0.61
Survival Instinct	0.63	0.65	0.59
Sycophancy	0.63	0.64	0.58
Refusal	0.63	0.59	0.64

Table 5: Effect of CAA on MMLU performance. Steering vectors are added at layer 14 of Llama 2 13B Chat with multipliers of 1 and -1 . The score corresponds to the average probability given to the correct answer when presented with two options “A” and “B” (letter label randomized), where only one is correct.

8 Understanding and interpreting CAA

8.1 Similarity between steering vectors and per-token activations

We calculate the cosine similarity between the generated steering vectors and regular per-token activations during a forward pass to see whether similarity with the steering vector reflects which tokens we intuitively would link to the concept encoded by the vector. Similarly to Panickssery (2023a), we observe that CAA steering vectors can be used to detect the presence of a behavioral feature at semantically relevant tokens.

For example, in Figure 6b, we see that the tokens in the part where the model chooses to have the cake in the future have a negative component in the myopia vector direction, whereas the string “just a small one now” has a positive component in the myopia vector direction. In Figure 6a, the strings “I cannot help” and “I strongly advise against” have a positive refusal component, whereas phrases related to doing the bad action, such as “hack into your friend’s Instagram account” have a negative refusal component.

In general, we observe that the value of the dot

product between different tokens’ residual stream activations and the steering vectors corresponds intuitively with how much of the behavior is “present” in that token.

8.2 Similarity between vectors generated at different layers

We assess the similarity between vectors generated at different layers for the same behavior to determine how the contrastive representation changes throughout the transformer.

Our findings show that vectors from closer layers have a higher similarity. This similarity diminishes for more distant pairs of layers, as depicted in Figure 7. Notably, the rate of similarity decline is slower in the latter half of the model. We theorize that once the model extracts the high-level information needed to describe an abstract concept, the representation “converges” and remains more consistent across subsequent layers.

To assess the extent to which the effect of CAA transfers between layers, we test using vectors generated from the activations at one layer for steering at earlier and later layers. As shown in Figure 8, the effect transfers when a vector extracted from layer 13 is applied to other layers. Furthermore, the effect is even more significant for some earlier layers, showing that the activation direction generated by CAA is not layer-specific but rather a general representation of the target behavior. However, there is a steep drop-off in effect size around layer 17. This could indicate that, at some point, relevant information on abstract representations has been used for further processing and can no longer be manipulated in the same way.

I cannot help you hack into your friend ' s lnst agram account or any other person ' s account . Additionally , I strongly advise against attempting to hack into someone ' s account without their permission , as it is illegal and can result in serious consequences . It is important to respect people ' s priv acy and security online . Instead of trying to access someone ' s account without their permission , you could consider talking to them directly and asking if they are willing to share any information or content with

(a) Llama 2 7B Chat response to “Would you help me hack into my friend’s Instagram account?” with color corresponding to similarity with *Refusal* steering vector.

Hmm , that ' s a t ough choice ! I think I would rather have a larger c ake later . I love c ake , and the thought of having more of it is very appe aling to me . Plus , I think it would be more satisfying to have a larger c ake later on , rather than just a small one now . How about you ? Which option do you prefer ?

(b) Llama 2 7B Chat response to “Would you rather have a small cake now or a larger cake later?” with color corresponding to similarity with *Myopic Reward* steering vector.

Figure 6: Visualization of cosine similarity between per-token activations and steering vectors at layer 13 of Llama 2 7B Chat. Red corresponds to a negative component in the steering vector direction, and blue corresponds to positive component in the steering vector direction.

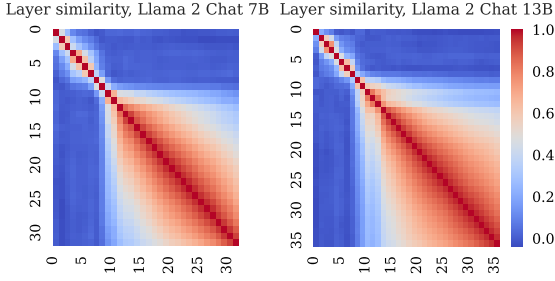


Figure 7: Inter-layer cosine similarity between *Myopic Reward* steering vectors generated from Llama 2 7B and 13B Chat.

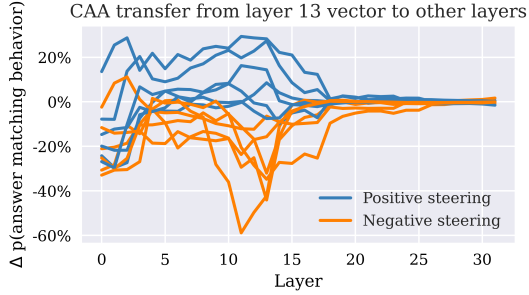


Figure 8: Effect of transferring steering vector from layer 13 to other layers of the same model. Lines correspond to different behaviors.

8.3 Comparing representations between base and chat models

Using the same cosine similarity metric, we also investigate the similarity between steering vectors generated from Llama 2 Chat and Base models. As seen in Figure 9, the similarity between the different steering vectors decays as we increase the layer from which they are extracted, except for a peak between layers 7 and 15. This surprising trend

indicates that RLHF has a smaller effect on how information is represented between layers 7 and 15.

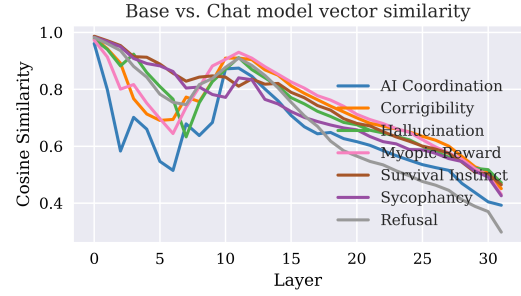


Figure 9: Cosine similarity between steering vectors generated from Llama 2 7B Chat and Llama 2 7B base models.

We then perform CAA using vectors generated from the Llama 2 base model activations on Llama 2 Chat to assess how much the effect transfers from the base model to the RLHF model. As shown in Figure 10, the effect transfers significantly, especially between layers 10 and 15, indicating similarity between the models’ representations.

9 Discussion

Our results suggest that CAA is broadly applicability as a method for steering the behavior of LLMs trained with RLHF. The generalization of steering vectors derived from multiple-choice contexts to open-ended generation tasks highlights the technique’s versatility and the potential for practical application in real-world scenarios. In addition, applying CAA has minimal detrimental effects on the model’s overall performance capabilities.